Parameterized Verification

Abstract

Parameterized verification is the problem of establishing that a correctness property holds for a family of systems indexed by a natural number n, representing the number of components, for all values of n simultaneously. The problem is undecidable in general, but a rich landscape of techniques has been developed that achieve decidability or effective automation for practically important classes of systems.

This chapter surveys the main approaches. Cutoff methods reduce verification to a finite number of instances by exploiting symmetry: a cutoff theorem asserts that correctness up to some bound k implies correctness for all sizes, and dynamic variants such as view abstraction determine the cutoff adaptively during the analysis. Regular model checking represents configurations as words and transition relations as finite-state transducers, reducing reachability to transducer closure; its lightweight variant, monotonic abstraction, uses upward-closed sets and backward reachability to obtain termination via well-quasi-ordering arguments. Abstraction and network invariant methods reduce the parameterized problem to a finite one by replacing the unbounded environment of a process with a single abstract summary. Deductive and proof-based methods construct inductive invariants that hold symbolically for all n, with automation provided by invisible-invariant synthesis and interactive proof assistants. SMT-based methods, including IC3/PDR with quantified invariant inference and array-based system encodings, automate invariant discovery using modern satisfiability solvers.

Two application domains illustrate how these techniques extend to harder settings. For concurrent programs under weak memory models, where hardware reordering introduces unbounded per-thread buffers, a dual load-buffer semantics restores the monotonicity required for backward reachability, giving the first decidability result for parameterized TSO verification and a PSPACE-completeness characterization; the richer Intel x86 model (eTSO) requires better quasi-orderings, marking their first use in program verification. For parameterized quantum circuits, where the parameter counts qubits rather than processes and correctness is a relational input-output property, synchronized weighted tree automata provide a symbolic representation of infinite families of quantum states, and a transducer-based composition framework yields the first fully automatic verification of parameterized quantum programs.

1 Introduction

Modern computing systems are increasingly built around the idea of replication: dozens of processors sharing a cache, hundreds of nodes collaborating in a distributed protocol, thousands of agents coordinating in a multi-agent system, or an arbitrary number of qubits processed by a quantum algorithm scaled to the

problem at hand. A fundamental question pervades all of these settings: how do we know that such systems behave correctly, regardless of how many components are involved?

Traditional formal verification techniques, such as model checking, work by exhaustively exploring the states of a system. They are highly effective when the system is fixed and finite, but become problematic as the system scales. When the number of components, processes, nodes, threads, agents, or qubits, is left as a free parameter, the state space is infinite, and classical finite-state methods no longer apply directly. This is the setting that **parameterized verification** addresses: proving or disproving that a correctness property holds for a *family* of systems, indexed by a natural number n representing the number of components, for all values of n simultaneously.

Properties and systems of interest. The correctness properties most commonly studied in parameterized verification are *safety* properties, which assert that something bad never happens (e.g., mutual exclusion: no two processes occupy a critical section at the same time), and *liveness* properties, which assert that something good eventually happens (e.g., every request is eventually served). Fairness conditions on scheduling, specifying how components take turns, are also of interest, particularly in combination with liveness. In the quantum setting, the relevant correctness notion is different in character: one typically asks for *relational* or *functional* properties, asserting that a parameterized circuit correctly maps every input state to the expected output state, for all input sizes simultaneously.

The range of systems to which parameterized verification applies is broad. Classic benchmarks include *mutual exclusion protocols* such as Burns', Peterson's, Dekker's, and Szymanski's algorithms, where the goal is to show that no two of an arbitrary number of processes can simultaneously enter their critical sections. *Cache coherence protocols*, such as the German and MESI protocols used in multiprocessor hardware, must maintain consistency of shared data across an unbounded number of caches. *Token-passing rings* pass a single token around a linear or circular array of processes, and safety amounts to showing that at most one process holds the token at any time. *Distributed consensus protocols*, such as variants of Paxos and leader election algorithms, must reach agreement among an arbitrary number of participants despite failures. Beyond these classic examples, parameterized verification applies to broadcast communication networks, producer-consumer systems, concurrent programs running on modern multiprocessor hardware under *weak memory models*, and, most recently, parameterized quantum circuits, where the number of qubits scales with the problem instance.

Fundamental difficulty. The challenge of parameterized verification is fundamental. In general, it is undecidable [AK86]: there is no algorithm that, given an arbitrary parameterized system and property, correctly determines whether the property holds for all system sizes. This result follows from the ability

to simulate the computation of a Turing machine using parameterized processes. The undecidability of the problem has not, however, prevented significant progress. Instead, it has motivated a rich landscape of techniques, each identifying meaningful and practically important subclasses of systems for which verification *is* tractable, and developing algorithms, abstractions, and proof methods tailored to those subclasses.

A unifying theme across many of the approaches covered here is the need for a *finite symbolic representation* of infinite families of system states. In the classical setting, regular languages and upward-closed sets play this role; in the quantum setting, *synchronized weighted tree automata* serve an analogous purpose. Another recurring theme is *monotonicity*: the observation that adding more components to a system can only enable more behavior, not less, is exploited in cutoff arguments, backward reachability algorithms, and well-structured transition system frameworks alike. A third theme is the tension between automation and expressiveness: fully automatic methods necessarily restrict the class of systems or properties they handle, while deductive and proof-based methods gain generality at the cost of requiring more human guidance.

Overview of approaches. This chapter covers the main lines of work in parameterized verification, organized around the following themes.

Early foundations (Section 2) trace the origins of the field to the first formulations of the parameterized model-checking problem: a control process together with an unbounded population of identical user processes communicating by rendez-vous, for which reasoning about the whole family reduces to a finite analysis [GS92]; automata used as inductive invariants for linear networks of arbitrary size [SG99]; and symmetry reduction for populations of identical processes [ES96, GS97].

Cutoff methods (Section 3) exploit the symmetry and homogeneity of parameterized systems to reduce verification to a finite number of instances. A *cutoff theorem* asserts that if a property holds for all system sizes up to some bound k, it holds for all sizes. Static cutoff theorems, due to Emerson, Namjoshi, and Kahlon, apply to token-passing rings, broadcast protocols, and systems with restricted communication [EN95, EK03]. *Dynamic cutoff methods*, exemplified by *view abstraction* [AHH13, AHH16], determine the cutoff adaptively during verification: they track what each process can *observe* about its neighbors, its *view*, and run simultaneous under- and over-approximation procedures that terminate as soon as either a bug or a safety proof is found.

Regular model checking (Section 4) takes a symbolic approach, representing configurations as words over a finite alphabet and sets of configurations as regular languages [KMM$^+$01, BW98]. Transitions are encoded as length-preserving finite-state transducers, and reachability analysis reduces to computing the transitive closure of a transducer. A lightweight variant, *monotonic abstraction*, replaces full regular-language representations with upward-closed sets, enabling efficient backward reachability anal-

ysis that terminates by a well-quasi-ordering argument [Abd10].

Abstraction and network invariants (Section 5) reduce the parameterized problem to a finite one by constructing a small abstract system that over-approximates all instances. The *network invariant* method represents the collective behavior of all processes except one by a single abstract process [WL89, PXZ02]. Finding the right invariant can be done manually, by abstract interpretation, or by counterexample-guided refinement.

Deductive and proof-based methods (Section 6) approach parameterized verification by constructing formal proofs of inductive invariants that hold for all system sizes simultaneously. Rooted in the foundational work of Owicki and Gries [OG76] and Lamport's TLA+ [Lam94], these methods carry out all reasoning symbolically over an arbitrary n, avoiding explicit enumeration. Recent work combines deductive reasoning with automated lemma generation and invariant synthesis to reduce the burden on the user.

SMT-based methods (Section 7) leverage modern Satisfiability Modulo Theories solvers to automate the search for inductive invariants. The IC3/PDR algorithm [Bra11], originally designed for finite-state hardware verification, has been extended to the parameterized setting through quantified invariant inference. Tools such as Ivy [PMP+16] let users specify systems in first-order logic and automatically derive invariants that certify safety for all n. Array-based system encodings [GR10] provide another SMT-friendly formalism and have seen success on distributed protocols including Paxos variants.

Applications to weak memory models (Section 8) represent a challenging application domain in which classical parameterized verification techniques must be adapted to handle the unbounded store or load buffers that arise from hardware memory reordering. The *dual semantics* for TSO makes the system monotone and amenable to WSTS-based backward reachability, yielding both the first decidability result and a practically efficient algorithm for parameterized TSO verification [AABN16, AABN18]. The precise complexity of this problem was subsequently pinned down to PSpace-completeness [AAR20]. The Intel x86 architecture implements a richer model (eTSO) whose parameterized verification requires the novel machinery of *better quasi-orderings* [AAB+26], marking the first use of BQOs to establish decidability in program verification.

Parameterized verification of quantum circuits (Section 9) is a recently opened frontier, in which the parameter n counts qubits rather than processes, and the correctness criterion is functional rather than safety-oriented. A new automata model, *synchronized weighted tree automata*, provides a symbolic representation of infinite families of quantum states, and a transducer-based framework handles parameterized gate semantics and circuit composition [ACH+25].

Scope and organization. The rest of this chapter is organized as follows. Sections 2–9 cover each of the approaches described above in detail, discussing the underlying ideas, key results, repre-

sentative algorithms, and important benchmarks. Section 10 discusses industrial applications of parameterized verification. Section 11 situates this chapter within the broader context of the book, drawing connections to related topics covered in other chapters. Throughout, we aim for an accessible presentation with minimal formalism, providing intuition and representative examples in preference to full technical detail. Pointers to primary sources are given for readers who wish to explore individual topics further.

2 Early Foundations

The questions studied in this chapter were first crystallized in a body of work from the late 1980s and 1990s that established both the model and the basic techniques of parameterized verification. A foundational formulation considers a system composed of a single control process together with an unbounded collection of user processes sharing a common definition, communicating through rendez-vous synchronization in the style of CCS [GS92]. The central contribution was to show that reasoning about the entire infinite family can be reduced to a finite analysis: decision procedures were given for checking whether every process satisfies a given indexed temporal specification, and the reachability questions arising in the analysis were connected to the theory of Petri nets (vector addition systems), whose coverability problem is decidable. This early demarcation of the boundary between decidable and undecidable cases, and of the algebraic structures underlying it, set the agenda for much of the work that followed.

A complementary early line of work targeted linear networks of arbitrary size and introduced automata as inductive invariants [SG99]. A correctness property is itself specified by an automaton, from which an invariant automaton is synthesized; an induction on the number of processes then establishes that the property is preserved as the network grows. The method was shown to be complete for safety properties. It directly anticipates both the network-invariant approach of Section 5 and the regular model checking framework of Section 4, in which sets of global configurations of unbounded width are represented symbolically by finite automata.

A further theme running through this early work is the exploitation of the symmetry inherent in a population of identical processes. Because the identity of an individual process is immaterial to the global behavior of the system, verification can be carried out over equivalence classes of configurations rather than over configurations themselves [ES96]. Combining this observation with fairness leads to on-the-fly model checking procedures that exploit symmetry to keep the explored state space manageable [GS97]. Taken together, these strands, finite reductions for synchronizing populations, automata-based invariants for linear networks, and symmetry reduction for replicated components, form the conceptual foundation on which the remainder of this chapter builds.

3 Cutoff Methods and Dynamic Cutoffs

One of the most elegant ideas in parameterized verification is the *cutoff theorem*. The key insight is that, for certain classes of systems, one does not need to verify infinitely many instances. Instead, it suffices to verify systems up to some finite bound, the cutoff, and correctness for all larger instances follows automatically [EN95, EK03].

Intuitively, this works because processes in many symmetric systems are interchangeable. If the behavior of a system with n processes can be faithfully represented by a system with at most k processes (the cutoff), then verifying the k-process system is enough. Cutoff results have been established for important classes such as token-passing rings, broadcast protocols, and systems where processes communicate in restricted ways. A significant body of work has developed cutoffs for mutual exclusion and cache coherence protocols [EK03].

However, classical cutoff theorems are *static*: the bound k is fixed ahead of time and must be established before verification begins. This can be limiting: the cutoff may be large, difficult to compute, or may not exist at all for systems with richer interaction patterns or more complex topologies.

Dynamic cutoff methods address this limitation by allowing the abstraction boundary to be determined adaptively, during the verification process itself, rather than fixed in advance. A prominent and well-developed technique in this family is **view abstraction** [AHH13, AHH16]. Rather than committing to a single small representative system upfront, the method reasons about what a process can *observe* about the rest of the system, and uses those local observations to construct an on-the-fly abstraction.

The starting point is to model the global state of an n-process system as a *configuration*: a sequence of local process states. A *view* of size k is a subsequence of length k, a projection of the global configuration onto a subset of k processes. Instead of tracking full configurations (whose length varies with n), the method tracks the set of all views of size up to k that are reachable across *any* instance of the system.

For a given precision parameter $k \in \mathbb{N}$, the framework runs two procedures in parallel. The first is an *under-approximation*: it performs concrete reachability analysis on instances of fixed size up to k, and reports a bug if a bad configuration is found. The second is an *over-approximation* based on abstract interpretation. It defines an abstraction function α_k, which maps a configuration to its set of subwords of size up to k, and a corresponding concretization γ_k, which maps a set of views back to all configurations whose subwords are all contained in that set. A fixpoint V_k is then computed over views using an abstract post operator. The key soundness property is that the set of reachable configurations R satisfies $R \subseteq \gamma_k(V_k)$ for all k, and the approximation becomes tighter as k grows: $\gamma_{k+1}(V_{k+1}) \subseteq \gamma_k(V_k)$.

A crucial implementation insight is that the abstract post operator can be computed using only concrete configurations of size $k{+}1$, rather than the full (infinite) concretization. This is be-

cause the witnesses required to enable a global transition consist of a single extra process; accommodating one additional process beyond the view size is always sufficient. This makes the fixpoint computation entirely finite and automatic.

If the over-approximation establishes $\gamma_k(V_k) \cap Bad = \emptyset$, the system is declared safe. If the under-approximation finds a reachable bad configuration, the system is declared unsafe. In either case, the current k is a *dynamic cutoff*, a point discovered during verification beyond which the search need not continue. If neither procedure is conclusive, the precision is increased by incrementing k and the analysis is repeated.

The procedure is guaranteed to terminate whenever the set of reachable configurations is *downward closed* with respect to the subword ordering on configurations. This condition holds in particular for *monotonic systems*: systems whose transition relation is monotone with respect to the subword ordering, in the sense that adding processes to a configuration can only enable more transitions, not fewer. This covers a broad and well-studied class of systems, including lossy channel systems, Petri nets (where the method solves the coverability problem), and timed Petri nets. For Burns' mutual exclusion protocol, for instance, the method terminates at $k = 2$: already with two-process views it can be shown that no bad configuration is reachable for any system size.

View abstraction has been successfully applied to a range of protocols, including mutual exclusion algorithms such as the fine-grained version of Szymanski's protocol, a benchmark that had not been verified automatically by other methods, as well as cache coherence protocols and distributed systems with complex process interactions. The framework extends naturally to other process topologies such as multisets, rings, and trees, and to settings where global transitions are not assumed to be atomic [AHH13].

Together, static cutoff theorems and dynamic approaches like view abstraction form a complementary toolkit: static cutoffs offer clean theoretical guarantees for well-structured protocol families, while dynamic methods provide greater flexibility and automation for protocols that fall outside those families.

4 Regular Model Checking

Regular Model Checking (RMC) is a symbolic framework for verifying parameterized and infinite-state systems by representing both sets of configurations and transition relations using the theory of regular languages and finite automata [KMM$^+$01, BW98]. The key idea is that regular sets serve as an efficient symbolic representation of infinite state spaces, playing a role analogous to that of Binary Decision Diagrams (BDDs) in symbolic model checking of finite-state systems [Abd10].

In its basic form, RMC represents a system as follows. A *configuration* is a finite word over a fixed alphabet Σ, where each symbol encodes the local state of one component (a process, a stack element, a channel message, etc.). Sets of configurations, including the initial set and the set of bad configurations, are represented as regular languages over Σ. The transition relation is encoded as a *length-preserving finite-state transducer* T over

pairs $(a, a') \in \Sigma \times \Sigma$: the transducer accepts a pair of words (w, w') if and only if the system can move from configuration w to configuration w' in one step.

This simple setup already captures a wide variety of models. Parameterized systems consisting of an arbitrary number of finite-state processes in a linear or ring topology can be modeled by letting each position in the word represent one process. Systems operating on unbounded queues, stacks, or integer variables can similarly be encoded by letting word positions represent data structure contents. The length-preserving constraint on the transducer is not a restriction for safety verification: for parameterized systems it simply means that the number of processes is fixed at the start of a run, while being arbitrary across runs.

Safety verification reduces to checking whether $(I \circ T^*) \cap B = \emptyset$, where I is the regular set of initial configurations, B is the regular set of bad configurations, and T^* is the reflexive transitive closure of the transducer relation. Liveness verification reduces to a *repeated reachability* problem: checking whether there exists an infinite computation from I through T that visits a set of accepting configurations F infinitely often.

The central computational challenge is computing T^+ (or $I \circ T^*$) effectively. Naively iterating the transducer, computing $T^1, T^2, T^3, \ldots$, does not converge in general. The standard approach is to use *acceleration* or *widening* techniques that jump directly to a representation of T^+. Acceleration is typically based on *quotienting*: one finds an equivalence relation $\simeq$ on the states of a finite approximation T^n, merges equivalent states to obtain a quotient transducer $T^n/\simeq$, and checks whether the result characterizes T^+. The difficulty is that quotienting generally enlarges the accepted language: $L(T^n) \subseteq L(T^n/\simeq)$, often strictly. A key theoretical contribution is the identification of equivalence relations that *preserve* the transitive closure, i.e., for which $(T/\simeq)^+$ is equivalent to T^+. Such relations can be derived from forward and backward bisimulations on the states of the *history transducer*, a canonical infinite transducer built as the union of all T^n for $n \geq 1$, where each state records the full sequence of T-states visited during product construction. Minimizing the history transducer via suitable bisimulations yields a finite quotient that characterizes T^+ [BJND03, AJNd02].

While the transducer-based approach is elegant and general, it can be computationally expensive and may not scale to larger protocols. An important line of work within the RMC framework addresses this by replacing the full power of regular languages with a lighter symbolic representation based on *upward-closed sets* [Abd10], in a technique known as *monotonic abstraction*. The approach targets parameterized systems with a linear process topology and global transitions (both universal and existential). Configurations are again words over the process state alphabet, and the subword ordering $c_1 \preceq c_2$ (meaning c_1 occurs as a not-necessarily-contiguous subsequence of c_2) plays a central role. Bad configurations are characterized by upward-closed sets with respect to this ordering: if a configuration is bad, then any larger configuration (with more processes) is also bad. Such sets

have a finite symbolic representation in terms of their minimal elements, making them far more tractable than arbitrary regular sets.

The core idea is to replace the concrete transition relation $\rightarrow$ with an abstract relation $\rightarrow_A$ that over-approximates $\rightarrow$ and is monotone with respect to $\preceq$. Local and existential transitions are already monotone and require no approximation. Universal transitions, where a process may move only if all processes in its context satisfy some condition, are *not* monotone, since adding more processes to a configuration may violate the universal condition and disable the transition. The abstraction handles this by allowing a configuration to silently drop processes that obstruct the transition, effectively making the universal condition vacuously satisfiable. Since $\rightarrow_A$ is an over-approximation, any safety property that holds in the abstract model is guaranteed to hold in the concrete one.

Safety verification then proceeds by *backward reachability analysis*. Starting from the upward-closed set Bad of bad configurations, one computes the sequence $U_0 = Bad$, $U_{i+1} = U_i \cup Pre_A(U_i)$, where Pre_A is the predecessor operator under $\rightarrow_A$. Because $\rightarrow_A$ is monotone, each U_i remains upward-closed and can be represented by a finite set of generators. Termination is guaranteed by Dickson's lemma: the subword ordering $\preceq$ is a well quasi-ordering on words, so every ascending chain of upward-closed sets stabilizes after finitely many steps. The algorithm terminates when no new generators (up to subsumption) are produced; if none of the generators represents an initial configuration, the system is safe.

RMC has been successfully applied to token-passing systems, mutual exclusion algorithms, and systems with unbounded communication channels, stacks, and integer variables. Beyond the linear word model, the framework extends to configurations represented as trees, enabling verification of systems with hierarchical or recursive structure [AJNd02]. Further extensions handle broadcast communication, binary rendezvous, processes operating on numerical variables with gap-order constraints, and systems running under weak memory models. The connection between monotonic abstraction and well-structured transition systems provides general decidability results for classes of systems that extend well beyond the purely parameterized setting [Abd10].

5 Abstraction and Network Invariants

Abstraction-based methods tackle the parameterized problem by reducing it to a finite, non-parameterized one. The idea is to construct an abstract system — typically a small, fixed-size system, that *over-approximates* the behavior of the full parameterized family. If a safety property holds in the abstract system, it holds in all concrete instances [WL89, PXZ02].

A particularly influential instantiation of this idea is the *network invariant* method [WL89], further developed in [PXZ02]. Consider a system consisting of n copies of a process template running in parallel. A network invariant is a single process I (or

small composition of processes) that captures the collective behavior of all $n-1$ processes that a given process interacts with. Verifying the property then reduces to checking the two-process composition of one concrete process with the invariant I, regardless of n. If this finite system satisfies the property, and I is a valid over-approximation of the environment, then all n-process instances satisfy it too.

Finding the right invariant automatically is the central challenge. Approaches range from manual construction guided by domain knowledge, to automated inference using abstract interpretation, to counterexample-guided abstraction refinement (CEGAR) adapted to the parameterized setting. In the CEGAR approach, one begins with a coarse invariant, checks the finite abstract system, and if a spurious counterexample is found, refines the invariant to rule it out. The work of [PXZ02] combines this idea with $(0, 1, \infty)$-*counter abstraction*, which tracks for each local state whether zero, one, or many processes occupy it. This finite abstraction is often coarse but captures enough structure to verify mutual exclusion and other coordination properties automatically, and the abstraction can be systematically refined when needed.

The network invariant approach is closely related to view abstraction (Section 3) in spirit: both methods replace the unbounded environment of a process by a finite abstract summary. The key difference is in how that summary is constructed and updated. Network invariants are defined globally as a fixed process before verification begins; view abstraction constructs the summary on the fly from the concrete reachability analysis, refining it incrementally as the precision parameter k increases.

6 Deductive and Proof-Based Methods

Deductive approaches shift the verification task from automated state-space exploration to the construction of formal proofs, often assisted by theorem provers or proof assistants. The idea is to find invariants, properties that hold at every reachable state of the system for every n, and prove that they are preserved by every system transition, regardless of the number of processes [OG76, Lam94]. Since the proof is carried out symbolically over an arbitrary n, this style of reasoning handles unbounded parameters without requiring enumeration of instances.

Classic work by Owicki and Gries [OG76] laid the groundwork for assertional reasoning about concurrent programs: each process is annotated with pre- and post-conditions, and an interference freedom check ensures that no process's execution invalidates another's assertions. The method was designed for a fixed, known number of processes, but the underlying proof discipline, inductive invariants plus non-interference, scales naturally to the parameterized case by carrying out the annotation and check for a symbolic, arbitrary process in a family. Lamport's TLA+ [Lam94] provides a formal language for specifying and proving properties of concurrent and distributed systems in this spirit, and has been applied to industrial-scale protocols at Amazon Web Services and Intel, among others.

A significant challenge for deductive methods is the discovery of inductive invariants: properties strong enough to be preserved under all transitions and to imply the desired safety property, but not so strong as to be unprovable. The *invisible invariants* technique of Pnueli, Ruah, and Zuck [PRZ01] addresses this by automatically generating candidate invariants from small finite instances of the system and then checking whether they hold inductively for all n. The idea is that the right invariant is often already visible in small instances and can be generalized to the full parameterized family. This makes deductive verification partly automatic, at the cost of completeness: if the right invariant is not among those generated, the procedure does not terminate with a proof.

More recent work integrates deductive reasoning with interactive proof assistants, allowing arbitrarily complex invariants to be stated and verified with machine-checked proofs. The Ivy system [PMP+16], which we return to in Section 7 in its SMT-based incarnation, is representative of this direction: it combines an interactive specification language with automated checking of proof obligations. Proof-based methods tend to be the most expressive and general of all approaches to parameterized verification, capable of handling protocols with complex data structures, arithmetic, and dynamic topology. Their principal limitation is the burden they place on the user: finding and expressing the right invariant requires deep insight into the protocol's correctness argument, and scaling to large or complex protocols remains demanding even with modern tool support.

7 SMT-Based Methods

The rise of powerful Satisfiability Modulo Theories (SMT) solvers over the past two decades has opened new avenues for parameterized verification. SMT-based methods encode system states, transitions, and properties as logical formulas and leverage highly optimized solvers to reason about them automatically [BSST09]. Their main advantage over purely deductive approaches is that the invariant search is largely automated; their main advantage over symbolic model checking methods is that the state space is represented logically rather than explicitly, making them applicable to systems with rich data and large process counts.

A prominent line of work in this space is *IC3/PDR* (Property Directed Reachability), originally developed for finite-state hardware verification [Bra11] and later extended to the parameterized setting through quantified invariant inference. IC3/PDR works by incrementally constructing an inductive strengthening of the safety property: it maintains a sequence of formulas that approximate the set of states reachable within increasing numbers of steps, using counterexamples to push these approximations forward and inductive checks to confirm that they constitute a valid invariant. In the parameterized setting, these formulas are universally quantified over process identifiers, and SMT reasoning over the resulting quantified theories replaces the finite propositional checks of the original algorithm.

Tools such as Ivy [PMP+16] and MYPYVY build on these ideas

to provide interactive and automated frameworks for specifying and verifying distributed protocols. Ivy allows users to specify systems in a first-order logical language with a decidable fragment (essentially, universal formulas over uninterpreted relations) and to interactively supply lemmas that guide the invariant synthesis. When the user identifies the right structure, the tool checks the proof obligations automatically using SMT. In practice, this workflow has successfully verified a range of distributed protocols, including variants of Paxos, two-phase commit, and leader election.

Another direction uses *array-based systems* [GR10], where the parameterized state is modeled as an array indexed by process identifiers, and quantified SMT formulas express properties over all processes simultaneously. The MCMT tool [GR10] implements model checking for array-based systems via backward reachability in a quantified fragment of first-order logic, using SMT solvers to discharge the fixpoint conditions. This approach handles systems in which each process has a local state drawn from an arbitrary (possibly infinite) domain, modeled as an interpreted theory in the SMT solver. Array-based methods have seen success on distributed protocols including Paxos variants and have connections to the theory of array property fragments studied in program verification.

The relationship between SMT-based methods and the other approaches in this chapter is instructive. SMT methods can be seen as automating the invariant discovery step that deductive methods (Section 6) leave to the user, while using a logical representation of state that subsumes the regular-language and upward-closed-set representations of Sections 4–5. The trade-off is decidability: the quantified theories used by SMT-based tools are generally undecidable, and the tools rely on heuristics, user interaction, or restriction to decidable fragments to terminate. This places SMT-based methods at the automation-expressiveness frontier: more automatic than purely deductive approaches, but less guaranteed to terminate than cutoff or monotonic abstraction methods.

8 Applications to Weak Memory Models

Modern multiprocessor architectures do not guarantee Sequential Consistency (SC) [Lam79]: for performance reasons, processors reorder memory operations in ways that are invisible to a single thread but observable between concurrently executing threads. The resulting *weak* (or *relaxed*) *memory models* (WMMs) include Total Store Order (TSO), which underlies x86 and SPARC processors, and Partial Store Order (PSO), among others. Verifying concurrent programs under such models is substantially harder than under SC, because each thread is equipped with a buffer that decouples writes from their visibility to other threads, making the state space infinite even for finite-state programs. Parameterized verification under WMMs is harder still: one must reason simultaneously about an unbounded number of threads, each carrying an unbounded buffer.

The connection to the techniques covered in earlier sections

is direct. Monotonic abstraction (Section 4) applies when the system's transition relation is monotone with respect to a suitable ordering on configurations. The key contribution of the weak-memory line of work is to identify the right semantics and the right ordering to make this apply, even though store and load buffers introduce data that naively breaks monotonicity.

The TSO model and the dual semantics. Under the classical TSO semantics, each thread is equipped with an unbounded perfect FIFO *store buffer*. When a thread writes to a variable, the write is appended to its store buffer rather than committed to memory immediately; values propagate to shared memory non-deterministically in FIFO order. A thread reading a variable first checks its own buffer for the most recent pending write, and falls back to main memory if none is found. Fence instructions force the buffer to drain before execution continues.

Decidability of safety verification under TSO for a fixed number of threads was established by encoding store buffers as lossy channels and applying the theory of well-structured transition systems (WSTS) [ABBM10]. However, these encodings are complex: each channel message must carry a full memory snapshot, and the reduction involves non-deterministic guessing that leads to significant state-space explosion in practice.

An alternative approach introduces a *dual* semantics for TSO [AABN16, AABN18]. Rather than store buffers, the dual model equips each thread with a *load buffer*: values propagate non-deterministically from shared memory into load buffers, and reads are served from the head of the load buffer rather than from memory directly. Writes, in contrast, are committed to memory immediately and tagged in the buffer as *own* messages. The dual semantics is provably equivalent to the classical one, the two models reach the same sets of local states, but offers significant practical advantages. Because load buffers behave as lossy channels in a straightforward sense, the WSTS framework applies with far less overhead than in the store-buffer encoding, yielding a simpler decidability proof and a more efficient verification algorithm.

The dual-semantics approach extends naturally to parameterized verification. In the parameterized setting, the number of threads is a free parameter, and one asks whether a bad local state is reachable for *any* number of threads. Parameterized configurations are ordered by a well-quasi-ordering that combines the subword ordering on load buffer contents with a matching on process identifiers; the transition system is shown to be monotone with respect to this ordering; and backward reachability over upward-closed sets of configurations terminates by Dickson's lemma. This yields the first decidability result for parameterized verification of concurrent programs under a weak memory model [AABN16, AABN18]. Moreover, experiments with a prototype tool show that parameterized verification is often *more* efficient than bounding the number of threads to a concrete value: the minor sets produced by the backward analysis are typically much smaller than the configuration sets generated by bounded-instance analyses, and the tool outperforms bounded-instance

tools by up to two orders of magnitude on standard mutual exclusion and memory-model benchmarks.

Complexity of parameterized TSO verification. While the dual-semantics approach establishes decidability, it leaves open the precise computational complexity of parameterized TSO verification. This question was settled in [AAR20], where it is shown that the reachability problem for a parameterized system of identical processes under TSO is PSPACE-complete.

The PSPACE upper bound is obtained by combining the load-buffer (dual) semantics with a careful observation about what information needs to be tracked: it suffices to maintain, for each process, the set of write operations it has issued together with the current value it observes for each variable, rather than the full buffer contents. This yields a polynomial-space abstraction that is both sound and complete for reachability. The matching lower bound is established by a reduction from the reachability problem for 1-safe Petri nets, and the reduction relies crucially on the ability of processes to read the *initial values* of variables from memory. This dependence on initial values is not merely an artefact of the proof: when processes are forbidden from reading initial memory values, the reachability problem drops to polynomial time, since the relevant information can be captured by a polynomial-sized abstraction tracking only the set of writes issued across all processes. A further variant is studied in which processes may perform *atomic read-modify-write* (RMW) operations; it is shown that adding RMW causes the complexity to jump to non-primitive recursive, matching the known hardness of reachability for general lossy channel systems. Together, these results give a precise and tight complexity picture for parameterized TSO verification across different fragments of the model.

The Intel x86 (eTSO) model. Real Intel x86 processors implement a memory model that goes beyond TSO. The formal model eTSO [RMV22] accounts for multiple memory types — write-back (wb), write-combining (wc), and strong-uncacheable, as well as non-temporal writes (ntw). Write-back writes behave as in TSO, while non-temporal writes on different variables may overtake one another, similarly to PSO. This makes eTSO strictly more relaxed than TSO, and the interaction between wb and ntw operations introduces subtle orderings that do not arise in either TSO or PSO alone. Reachability under eTSO for a fixed number of threads is in general undecidable, but decidability can be recovered by bounding the number of *alternations* between wb and ntw write stages per thread [AAB$^+$24]. A run is k-alternation bounded if the writes of each thread can be partitioned into at most k blocks, alternating between blocks containing only wb writes and blocks containing only ntw writes. Even under this restriction, executions may involve unbounded sequences of operations within each block, and complex shuffles of writes across blocks are permitted.

Parameterized verification of this k-alternation bounded problem was established in [AAB$^+$26]. The proof introduces a new

transition system, called the *NTWLoad* system, which combines the load-buffer technique from the dual TSO semantics (to handle wb writes while preserving monotonicity) with per-variable store buffers for ntw writes. Each eTSO thread is simulated by $2k$ concurrent *stage threads*, alternating between wb and ntw stages, with a finite interface that enforces the required causal dependencies across stages.

A central technical difficulty is that TSO store buffers are not monotone with respect to the standard subword ordering on words: inserting additional write messages can block reads that were previously enabled, breaking the simulation argument. This difficulty is circumvented by switching wb writes to the load-buffer representation, in which monotonicity is restored, while ntw writes retain their store-buffer treatment. The causal consistency across stages is maintained through an explicit bookkeeping mechanism that tracks obligations, speculated write values and expected termination labels for each stage, that must be discharged during execution.

This bookkeeping mechanism introduces a further complication: the resulting ordering on configurations is not merely a well-quasi-ordering (WQO) but must be a *better quasi-ordering* (BQO). BQOs are strictly stronger than WQOs and enjoy additional closure properties, in particular under the powerset construction, that WQOs do not. The ordering on NTWLoad configurations is built by combining a subword ordering on buffer contents with an equality constraint on stage information and a powerset-style matching on thread sets; showing this composite ordering is a BQO requires appealing to the closure of BQOs under products, word embeddings, and powerset operations. The transition relation is then shown to be monotone with respect to this BQO, placing the system within the WSTS framework and yielding decidability of backward reachability. To the best of our knowledge, this is the first application of BQO-based well-structured systems to prove decidability of a program verification problem [AAB$^+$26].

9 Parameterized Verification of Quantum Circuits

The preceding sections have developed parameterized verification in the classical setting: processes communicate through shared memory or message passing, the parameter n counts the number of processes, and correctness is expressed as a safety or liveness property over an infinite family of transition systems. A structurally analogous challenge arises in quantum computing. Most useful quantum algorithms, Shor's factoring algorithm, Grover's search, the Quantum Fourier Transform, are *parameterized by input size*: a quantum *program* takes n as input and dynamically generates the corresponding n-qubit circuit. Verifying such a program requires establishing correctness for all values of n simultaneously, a goal that parallels classical parameterized verification but requires substantially different technical machinery.

The verification problem. The correctness criterion for quantum programs is characteristically different from the safety and liveness properties of Sections 2–8. Rather than asking whether a bad state is ever reached, one asks whether the program computes the right function: given a set of quantum state pairs (p, q), does the program, for every input size n, correctly map input state p to output state q? This *relational* (or *functional*) specification captures input-output correctness across all parameterized instances simultaneously. It also subsumes *equivalence checking* between two parameterized programs — a property of direct practical importance for verifying that an optimized circuit implementation is functionally identical to a reference specification, regardless of n.

Existing approaches to quantum program verification have made notable progress but fall short of fully automatic treatment of general parameterized circuits. The Qbricks project [CBB+21] develops a deductive framework for quantum programs with rich specifications, but verification is not fully automated and requires user-provided proof annotations. The AutoQ tool [ACH+23, ACH+24] supports automated verification of quantum circuits using tree automata but is limited to a restricted class of parameterized instances. A fully automatic framework for general parameterized quantum programs was an open problem until recently [ACH+25].

Representing infinite families of quantum states. The central challenge in all parameterized verification is finding a finite symbolic representation of an infinite family of system states, one state space for each value of n. In the classical setting, regular languages over a finite alphabet serve this role (Section 4): a single finite automaton compactly represents all reachable configurations across all instance sizes. In the quantum setting the objects are quantum states rather than process configurations. An n-qubit quantum state is a vector of 2^n complex amplitudes, and a parameterized algorithm generates an infinite family of such vectors, one for each n, with a structured recursive form that grows exponentially in size but must be captured in a finite representation.

The framework of [ACH+25] achieves this through a new automata model called *synchronized weighted tree automata* (SWTAs). The starting point is the observation that an n-qubit quantum state can be naturally represented as a binary tree of depth n: each internal node corresponds to a qubit, and each leaf stores the complex amplitude associated with a particular computational basis state. Standard tree automata recognize sets of such trees, providing a symbolic representation of sets of quantum states that is analogous to finite automata over words. SWTAs extend this with two additional features. First, *colors* attached to tree nodes serve as synchronization labels, allowing the automaton to correlate corresponding positions across two different state trees; this is the mechanism by which input-output pairs are linked. Second, *weights* on transitions encode complex amplitudes, enabling the automaton to represent not merely which quantum states are possible, but what amplitude each state carries. Together, these

extensions yield a model that defines a *function* from color sequences to quantum states, rather than simply a set of states. This functional view is essential for relational verification: it allows a specification to assert what output state is expected for each input state, across all input sizes simultaneously.

Transducers for quantum gate semantics. In regular model checking, the transition relation of a parameterized system is encoded as a finite-state transducer, and reachability reduces to computing the transitive closure of that transducer (Section 4). The quantum analogue replaces the transition relation with a *quantum gate*, and the transducer must encode the linear transformation that the gate applies to the quantum state tree. The framework of [ACH+25] introduces a class of transducers that operate directly on SWTAs: given an input SWTA representing a family of quantum states, applying a gate transducer produces an output SWTA representing the family of states after the gate has been applied.

A central concern in any symbolic framework for quantum computation is *succinctness*: a quantum gate on n qubits is a $2^n \times 2^n$ unitary matrix, and a naive representation is exponential in n. The transducer representation avoids this blow-up. As an illustration, the transducer for the Quantum Fourier Transform on n qubits has a number of states that is *polynomial* in n, exploiting the recursive structure of the QFT to obtain an exponentially more compact representation than a matrix encoding would require. This mirrors the advantage of transducers over explicit transition tables in classical regular model checking.

A further key ingredient is a *composition operator* for transducers. Given transducers for two sub-circuits, the operator automatically constructs a transducer for their sequential composition. Starting from transducers for individual gates, repeated composition assembles the transducer for an entire circuit without user intervention. The operator extends to *parameterized programs*: given a transducer for a finite circuit constituting one layer of a parameterized algorithm, composition generates a transducer for the full parameterized version with an arbitrary number of such layers. This step, lifting from finite-circuit verification to parameterized verification, is the direct quantum analogue of computing the transitive closure of a transducer in classical regular model checking.

Verification algorithms and experimental results. With SWTAs as the specification formalism and transducers as the program model, functional verification reduces to checking *SWTA inclusion*: given the SWTA $\mathcal{A}$ specifying the desired input-output relation and the SWTA $\mathcal{A}'$ obtained by applying the program transducer to the input specification, verify that $[\![\mathcal{A}']\!] \subseteq [\![\mathcal{B}]\!]$. Equivalence checking between two parameterized programs reduces to an equality check between the output SWTAs they produce when applied to a complete set of linearly independent inputs; correctness of this reduction follows from the fact that quantum programs implement linear transformations, so agree-

ment on a complete input basis implies agreement on all inputs. Both inclusion and equality of SWTAs are decidable, and the framework provides practical algorithms for both.

The implementation, called AutoQ-Para [ACH+25], was evaluated on a diverse set of benchmarks spanning algorithmic, arithmetic, error-correction, and simulation circuits. It successfully verified the Bernstein-Vazirani algorithm, a ripple-carry adder, a syndrome extraction circuit from a quantum error-correcting code, and optimized implementations of Grover's search and Hamiltonian simulation, comparing them against their unoptimized reference counterparts. Verification times ranged from 14 milliseconds to 11 seconds, and to the best of the authors' knowledge these are the first fully automatic parameterized verifications of circuits of this kind.

Connections to classical parameterized verification. The structural parallels with classical parameterized verification are instructive. In both settings the parameter n indexes an infinite family of systems, and the goal is to verify a property of the entire family from a finite symbolic representation. Regular languages and SWTAs both fill this role compactly, words for classical configurations and trees for quantum states. Transducers encode transition relations in the classical case and gate semantics in the quantum case, and computing transitive closures or repeated compositions is the operation that lifts finite reasoning to the parameterized family in both. The SWTA inclusion check mirrors the backward reachability pattern of monotonic abstraction, amounting to verifying that no reachable output state lies outside the specification. What distinguishes the quantum setting is the algebraic structure of the state space: quantum states carry complex amplitudes subject to normalization and unitarity constraints, and gates are unitary linear transformations rather than arbitrary relations. The SWTA model is designed to respect this structure, with weights on transitions as the mechanism for tracking amplitudes within an automata-theoretic framework. The result is, in effect, a quantum lifting of regular model checking, adapted to the linear-algebraic semantics of quantum computation.

10 Industrial Applications

11 Relations to Other Chapters

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